Roll No. | | | | | | | |

End Semester Examination, December 2024

Name of the Course: B.Tech Biotechnology
Name of the Paper: Biochemistry

Semester: III

Paper Code: TBT 304

Time: 3Hours

Maximum Marks: 100

Note:-
(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1	(20marks)	
(a)	Discuss the Urea cycle for metabolism of alpha amino group of amino acids.	CO1
(b)	Draw structure of following amino acids (i) Valine (ii) Tyrptophan (iii) Cysteine (iv) Histidine (v) Glutamate	
(c)	Describe the possible fate of a glucose molecule undergo during aerobic conditions inside a living cell.	
Q2	(20marks)	
(a)	Differentiate between following terms (i) Anomers and epimers (ii) Disaccharides and polysaccharides (iii) Reducing and non-reducing sugars (iv) Starch and glycogen	CO2
(b)	Draw structures of following carbohydrates (i) α-D-Glucose (ii) D-Fructose (iii) Sucrose (iv) D-Galactose	
(c)	Explain the TCA cycle with all intermediates and enzymes involved in the pathway.	
Q3	(20marks)	

(a)	(i) Explain the factors on which melting point of a fatty acid depends. (ii) What are triacylglycerols? Draw neat diagram of cholesterol.	CO3
(b)	Discuss the HMP pathway for a glucose molecule and its significance in the cell.	
(c)	Calculate the number of ATP molecules produced during beta-oxidation of Palmitoyl-CoA fatty acid.?	
Q4	(20marks)	CO4
(a)	Differentiate in the following terms: (i) Active site and binding site of enzyme (ii) Hydrophilic and hydrophobic amino acids (iii) Metalloenzyme and metal activated enzymes. (iv) Cofactor and prosthetic group.	
(b)	Discuss Lock and key theory of enzyme activity and its challenges.	
(c)	Discuss the factors that governs the enzyme activity in a enzyme catalyzed reactions.	
Q5	(20marks)	CO5
(a)	Differentiate between following terms: (i) DNA and RNA (ii) Purine and pyrimidine (iii) Type A and B DNA (iv) Salvage pathway and De Nova pathway of nucleotide synthesis.	
(b)	Explain Waston crick model of DNA with a neat diagram.	
(c)	Explain chemiosmotic theory for the ATP production in a eukaryotic cell.	